

Brain-Controlled Home Automation using EEG and ML

1. Introduction

This project enables the control of home appliances (like LED or fan) just by thinking, using EEG signals. We use a low-cost EEG sensor, an ESP32, and a PC to process the brain signals with Machine Learning.

2. Objective

To create a mind-controlled home automation system where appliances are turned ON or OFF based on the user's thoughts, without any physical movement or voice commands.

3. Hardware Used

- BioAmp EXG Pill (EEG Signal Acquisition)
- ESP32 Microcontroller
- LED or Relay-Controlled Fan
- Jumper Wires
- USB cable for power and serial communication

4. Wiring and Connections

The connections are as follows:

- BioAmp EXG Pill V+ to forehead (via electrode)
- BioAmp EXG Pill V- to behind the ear (via electrode)
- BioAmp EXG Pill GND to reference (earlobe or wrist)
- BioAmp TX to ESP32 RX (GPIO 16 recommended)
- ESP32 GND to BioAmp GND
- Relay IN to ESP32 GPIO 23
- Relay VCC to 5V, GND to GND
- Appliance connected to Relay (NO and COM)

5. Working Principle

1. EEG signals are collected using the BioAmp EXG Pill and sent to a PC.
2. The PC runs a Python program to process and classify the signals using a trained ML model.

3. Based on the detected thought (ON/OFF), a command is sent from the PC to the ESP32.
4. The ESP32 receives the command and turns ON or OFF the appliance accordingly.

6. Machine Learning Integration

- EEG data is collected while the user thinks 'Turn ON' or 'Turn OFF'.
- The data is labeled and stored in a CSV file.
- A model (e.g., SVM, KNN) is trained to classify new incoming signals.
- The trained model runs in real-time to predict the user's intent.
- The command is sent to the ESP32 over Serial or Wi-Fi.

7. Budget

Part name	Link	Price
BioAmp EXG Pill	https://robu.in/product/bioamp-exg-pill-unassembled/	₹ 2,561.00
Brain BioAmp Band (6 Channel)	https://robu.in/product/brain-bioamp-band-6-channel/	₹ 629.00
38Pin CP2102 ESP-32 Wifi+Bluetooth Development Board with Type-C USB Interface	https://robu.in/product/38pin-cp2102-esp-32-wifiblueetooth-development-board-with-type-c-usb-interface/	₹ 456.00
Miscellaneous		₹371
Total		₹4,000

8. Conclusion

This project demonstrates how EEG signals can be used to control devices through thought alone, paving the way for hands-free smart home automation, especially beneficial for differently-abled users.